Dylan Pentland

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Education

Harvard 2022–2027 (expected)
Ph.D in Mathematics, Advised by Mark Kisin

MIT 2018–2022
B.S. in Mathematics 5.0/5.0 GPA

Preprints

A prismatic Riemann-Hilbert correspondence 2026
Shows an equivalence of categories between constructible p-Kummer-étale $\mathbf{Z}_p$-sheaves on the log adic generic fiber $(\mathrm{X}, \mathcal{M})_\eta$ for the logarithmic stratification induced by D and finitely generated unit Laurent crystals on the log prismatic site. I prove p^+-perverse t-exactness, give a local description of finitely generated unit crystals on a framed log affine in terms of (φ, Γ)-modules, and show pushforward compatibility for strict log smooth proper morphisms. Can be accessed here.

Appendix 2 to "A remark on an integral structure of the imperfect coefficient ring of (φ, Γ)-modules" by Takumi Watanabe 2026
This appendix explains for certain tamely ramified $K = \mathbf{Q}_p(p^{1/e})$ the non-existence of a prism $(\mathbf{A}_K^+, \varphi^\ell([p]_q))$ satisfying certain reasonable properties, namely that there is a Frobenius equivariant identification $\mathbf{A}_K^+[1/\varphi^\ell([p]_q)]_p^\wedge \simeq \mathbf{A}_K$ and moreover that $\mathbf{A}_K^+/p \simeq \mathrm{E}_K^+ \subset \mathrm{E}_K$ through this identification. Here $\mathbf{A}_K$ is Fontaine's imperfect period ring and E_K^+ the ring of integers of the field of norms of the cyclotomic tower of K. Can be accessed here.

Syntomification and crystalline local systems 2025
Extends a result of Bhatt-Lurie showing there is an equivalence of categories between reflexive sheaves on the stack $\mathbf{Z}_p^{\mathrm{Syn}}$ and $\mathbf{Z}_p$-lattices in crystalline representations to the general case of smooth $\mathrm{X}/\mathrm{Spf}\mathcal{O}_K$, where $K/\mathbf{Q}_p$ is a finite extension. I show $\mathsf{Perf}(\mathrm{X}^{\mathrm{Syn}})[1/p]$ is equivalent to a derived category of admissible filtered F-isocrystals when X is smooth proper. The key input is a calculation of syntomic cohomology with coefficients over $\mathcal{O}_K$ via an explicit syntomic complex. Submitted to *Algebra and Number Theory*. Can be accessed here.

Awards

NSF Fellowship 2022
Regeneron STS Finalist 2018

Research Talks

A prismatic Riemann-Hilbert correspondence November 2026
Boston University Number Theory Seminar

A prismatic Riemann-Hilbert correspondence 2026
University of Virginia Ramanujan-Serre seminar

Syntomification and crystalline local systems 2025
Harvard Number Theorists' seminar

Expository Talks

These are expository talks I gave at Harvard internal seminars.

Canonical modifications of Breuil-Kisin modules 2026
Harvard Number Theorists' seminar

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Lütkebohmert's Riemann Existence theorem 2026
p-adic Seminar

Mumford Curves and Jacobians 2025
Trivial Notions Seminar

The Emerton-Kisin correspondence 2025
p-adic Seminar

F-gauges and crystalline Galois representations 2025
Learning seminar on prismatic F-gauges

The Klein Quartic 2024
Trivial Notions Seminar

Derived prismatic cohomology 2024
Learning seminar on prismatic cohomology

The arc topology 2024
Learning seminar on prismatic cohomology

Twistor $\mathbf{P}^1$ 2023
Trivial Notions Seminar

Almost purity 2023
Learning seminar on prismatic cohomology

The Beilinson Fiber square 2023
Zygotop seminar

Shtukas 2023
Geometric Langlands support seminar

Nearby cycles 2023
Geometric Langlands support seminar

Honda-Tate theory 2022
Trivial Notions Seminar

Teaching and other activities

Linear Algebra 2025 Spring
Taught Math 21b at Harvard.

Summer Tutorial 2023 Summer
Taught a Harvard summer tutorial on Bass-Serre theory.

DRP Mentor 2023–
Mentored undergraduate students in étale cohomology, p-adic Hodge theory, rational points on elliptic curves, group theory, analytic number theory.

Head Counselor at PROMYS 2022
Managed program for motivated high school students learning number theory.

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List of Publications

Extensions of mod p representations of local division algebras 2024

With Andrew Keisling.
Journal de Théorie des Nombres de Bordeaux

Filtrations on block subalgebras of restricted universal enveloping algebras 2022

With Andrei Ionov.
Journal of Algebra and Applications

Computing L-polynomials of Picard curves in polylogarithmic time 2022

With Sualeh Asif and Francesc Fité.
Mathematics of Computation

Coefficients of Gaussian Polynomials Modulo N 2020

Electronic Journal of Combinatorics
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